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Modelling, simulation, and experimental validation for toluene removal from gas phase in a fixed bed adsorption column

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ABSTRACT

In this study, an adsorption technique was explored for the removal of toluene present in the gas phase by taking granular activated carbon as an adsorbent in a fixed bed adsorption column. Breakthrough curves were experimentally found out at varying bed height (0.015–0.025 m) and inlet concentration (7000–11500 ppm). Modelling and simulation studies were subsequently carried out to enable the comparison between predicted and experimental breakthrough curves. Here orthogonal collocation technique was employed to solve the model equations by discretising non-linear coupled partial differential equations into set of ordinary differential equations which were solved by using ODE15s solver of Matlab. The discretisation procedure has been described in detail. Finally, the flow rate was found out to be the most sensitive parameter after carrying out sensitivity analysis.

KEYWORDS

Adsorption; granular activated carbon; VOC; toluene; breakthrough curve

Symbols

C_b	toluene concentration in the bulk gas phase (kg m^{-3})
C_{bo}	initial toluene concentration in the bulk gas phase (kg m^{-3})
C_p	toluene concentration inside the pores (kg m^{-3})
C_{ps}	toluene concentration at the outer surface of the adsorbent (kg m^{-3})
D	combined diffusivity ($\text{m}^2 \text{s}^{-1}$)
D_{AB}	molecular diffusivity ($\text{m}^2 \text{s}^{-1}$)
D_k	knudsen diffusivity ($\text{m}^2 \text{s}^{-1}$)
D_L	axial dispersion coefficient ($\text{m}^2 \text{s}^{-1}$)
D_e	effective diffusivity ($\text{m}^2 \text{s}^{-1}$)
D_p	diameter of the adsorbent particle (m)
K	Langmuir isotherm constant ($\text{m}^3 \text{kg}^{-1}$)
K	Boltzmann's constant (J K^{-1})
k_m	external-film mass-transfer coefficient (m s^{-1})
L	length of the fixed bed (m)
M_A	molecular weight of toluene (kg kmol^{-1})
M_B	molecular weight of nitrogen (kg kmol^{-1})
P	pressure (Pa)
Q	volumetric flow rate (mL min^{-1})
q_m	maximum adsorption capacity of the adsorbent (kg kg^{-1})
R_p	radius of the adsorbent particle (m)
r_{AB}	molecular separation of collision, nm
r_p	Pore radius (m)
T	absolute temperature (K)
V_z	interstitial velocity (m s^{-1})

Greek letters

ε_b	bed porosity (–)
ε_{AB}	energy of molecular attraction (J)
α	particle porosity (–)
ρ	particle density (kg m^{-3})
μ	gas viscosity ($\text{kg m}^{-1} \text{s}^{-1}$)
ρ_g	gas density (kg m^{-3})
φ	shape factor (–)
τ	tortuosity factor (–)

Introduction

Rapid urbanisation, due to population growth and swift industrialisation has been a potential threat to the atmosphere that inspires researchers to think innovatively to deal with this colossal ongoing air pollution problem. Instances of smog and health problems have been increasing for last so many years owing to poor urban air quality. Lot of new air pollutants are being recognised and one such class of hazardous air pollutants are volatile organic compounds (VOCs), whose vapour pressure exceeds 0.5 kPa at room temperature [1]. Process industries are reported to be the principal source of VOC emission [2], whopping 230 million tons of VOCs are emitted from various sources which is a cause of concern [3]. Also, the adverse health effects of VOCs include eye, throat and nose irritation. While cancer, kidney and liver damage, etc. are some of the chronic effects [4]. The ill effects of VOCs on health and environment even at low concentrations calls for its immediate removal by employing a technique which is both environment friendly and less energy intensive. One such technique – adsorption has been used in the present work for the removal of VOCs.

The current work aims at removal of toluene (VOC) from the gas stream in a fixed bed adsorption column. The performance of a fixed bed adsorption column is always adjudged by (i) type of adsorbent used (greater surface area is desirable) and (ii) breakthrough time (longer times are desirable). This work employs granular activated carbon (commercial) as an adsorbent, chosen on the basis of literature review which revealed the use of carbon in numerous forms namely monolithic carbon aerogels [5], activated carbons [6], pecan-shell and almond shell based granular activated carbons [7], carbon-based nanocomposites [8], activated carbon/metal oxide composites [9], sugarcane bagasse derived porous carbons [10], coconut shell activated carbon [11], porous carbons from silica-rich rice husk char [12], spherical activated carbons [13].

Besides experimental investigation, mathematical model was also formulated in the present study for fixed-bed adsorption column to obtain breakthrough curves at varying operating conditions. Generally, two types of mathematical models have been employed for predicting breakthrough curves in fixed bed adsorption column i.e. linear driving force model (LDF) [14–18] and pore diffusion model [19–21]. LDF is a simplified model where the rate of adsorption is assumed to be linearly proportional to driving force. Here, the pore diffusion model has been employed which incorporated external mass transfer, internal diffusion and adsorption on the pores of the adsorbent. Set of partial differential equations thus obtained from the mathematical model were solved by using orthogonal collocation technique. The advantage of using orthogonal collocation technique over finite difference technique (most frequently employed by various researchers) is relatively lesser computational time on account of lesser number of ordinary differential equations formed following discretisation of partial differential equations. Step-wise discretisation procedure has also been described in this paper for the better understanding of the process. Subsequently, simulation studies and sensitivity analysis have been done to ascertain the most sensitive parameter with respect to breakthrough time.

The objectives of the present study are: (i) To conduct fixed – bed experiments for the removal of VOCs at varying bed heights and inlet concentrations. (ii) Modelling and simulation studies to envisage breakthrough curves under varying operating conditions. (iii) Sensitivity analysis.

Development of a mathematical model

Mathematical model has been developed in this section to envisage the concentration of VOC at the exit of the column with respect to time. Internal and external mass transfer resistances (characterised by D_e and k_m), non-ideal plug flow (characterised by D_L), instantaneous local adsorption equilibrium on the surface of the pore, spherical shape of the adsorbent particle, and isothermal conditions were taken into consideration while developing the mathematical model [22].

The non-dimensional VOC mass balance in the bulk phase along with boundary conditions is given by:

$$\left(\frac{\partial C_b^*}{\partial t}\right) = \frac{-1}{t_{\text{res}}} \left(\frac{\partial C_b^*}{\partial z^*}\right) + P \left(\frac{\partial^2 C_b^*}{\partial z^{*2}}\right) - Z(C_b^* - C_{ps}^*) \quad (1)$$

$$\text{where } t_{\text{res}} = \frac{L}{V_z}, \quad P = \frac{D_L}{L^2} \quad \text{and} \quad Z = \frac{3k_m(1 - \epsilon_b)}{R_p \epsilon_b}$$

$$C_b^* = \frac{C_b}{C_{b0}}$$

$$z^* = \frac{z}{L}$$

$$\text{and } C_{ps}^* = \frac{C_{ps}}{C_{bo}}$$

Boundary conditions:

- i at $t = 0$ and at all z^* , $C_b^* = 0$,
- ii at $t > 0$ and at $z^* = 0$, $C_b^* = 1$, and
- iii at $z^* = 1$, $\frac{\partial C_b^*}{\partial z^*} = 0$

The VOC mass balance inside the pores of the adsorbent is given by:

$$\left(\frac{\partial C_p}{\partial t}\right) + \left(\frac{1 - \alpha}{\alpha}\right) \rho \left(\frac{\partial q}{\partial t}\right) = D_e \left[\frac{\partial^2 C_p}{\partial r^2} + \frac{2}{r} \frac{\partial C_p}{\partial r} \right] \quad (2)$$

Assuming instantaneous equilibrium

$$\frac{\partial q}{\partial t} = \frac{\partial C_p}{\partial t} \frac{\partial q}{\partial C_p} \quad (3)$$

Here Langmuir isotherm was employed that provided a relationship between VOC adsorbed and its concentration in the pores of the adsorbent and is given by the following equation.

$$q = \frac{q_m K C_p}{1 + K C_p} \quad (4)$$

where q_m is the maximum adsorption capacity of the adsorbent in kg/kg and K is the Langmuir isotherm constant in m^3/kg .

Differentiating Equation (4) w.r.t C_p , we get

$$\frac{\partial q}{\partial C_p} = \frac{q_m K}{(1 + K C_p)^2} \quad (5)$$

From Equations (2)–(5), we get

$$\left(\frac{\partial C_p}{\partial t}\right) = \left[\frac{1}{1 + \rho \left(\frac{1 - \alpha}{\alpha} \right) \frac{q_m K}{(1 + K C_p)^2}} \right] D_e \left[\frac{\partial^2 C_p}{\partial r^2} + \frac{2}{r} \frac{\partial C_p}{\partial r} \right] \quad (6)$$

Now, the non-dimensional VOC mass balance inside the pores of the adsorbent along with boundary conditions is given by:

$$\frac{\partial C_p^*}{\partial t} = \frac{T \alpha (1 + K C_{b0} C_p^*)^2}{\alpha (1 + K C_{b0} C_p^*)^2 + \rho (1 - \alpha) q_m K} \left[\frac{\partial^2 C_p^*}{\partial r^{*2}} + \frac{2}{r^*} \frac{\partial C_p^*}{\partial r^*} \right] \quad (7)$$

where $T = \frac{D_e}{R_p^2}$

$$C_p^* = \frac{C_p}{C_{b0}}$$

$$\text{and } r^* = \frac{r}{R_p}$$

Boundary conditions:

- i at $t = 0$, $C_p^* = 0$,
- ii at $r^* = 0$, $\frac{\partial C_p^*}{\partial r^*} = 0$, and
- iii at $r^* = 1$, $\frac{\partial C_p^*}{\partial r^*} = R(C_b^* - C_{ps}^*)$ where $R = \frac{k_m R_p}{D_e}$

During the simulation, breakthrough curves were found out under varying conditions by using q_m and K as adjusted parameter models.

Solution of equations

The governing Equations (1) and (7) obtained in the previous section are non-linear coupled partial differential equations. Various techniques have been successfully used by different authors for the solving the set of PDE's like volume finite method [23], orthogonal collocation [24,25], finite difference [26–28], centred finite difference (MOL) [29], orthogonal collocation on finite elements [30] and finite element method [31]. Here orthogonal collocation technique was used to solve the governing equations and in the succeeding paragraph this technique has been described in detail for solving the resulting PDE's.

Orthogonal collocation

It is a very popular technique for solving non-linear partial differential equations. Here 12 collocation points have been taken along the length of the column between 0 and 1 i.e. 0.00921968, 0.0479413, 0.115048, 0.206341, 0.316084, 0.43738, 0.56261, 0.683915, 0.793658, 0.884951, 0.95205, and 0.9907803 and six collocation points i.e. 0.21535, 0.420638, 0.60625, 0.76351, 0.88508, and 0.96524 have been taken along the radial position of the adsorbent particle between 0 and 1 [32]. The following expressions are used for finding the first and second derivatives of $\emptyset$ along the length

of the bed.

$$\left. \frac{d\vartheta(t)}{dx} \right|_{x_i} = \sum_{j=1}^{N+2} A_{ij} \vartheta(x_j, t) \quad (8)$$

$$\left. \frac{d^2\vartheta(t)}{dx^2} \right|_{x_i} = \sum_{j=1}^{N+2} B_{ij} \vartheta(x_j, t) \quad (9)$$

where N is the number of internal collocation points (i.e. 12) and A and B are the matrices of size 14×14 .

Similarly, the expressions used for finding the first and second derivatives of ϑ along the radial position of the adsorbent particle are given by

$$\left. \frac{d\vartheta(t)}{dx} \right|_{x_i} = \sum_{j=1}^{N+1} A'_{ij} \vartheta(x_j, t) \quad (10)$$

$$\left. \frac{d^2\vartheta(t)}{dx^2} \right|_{x_i} = \sum_{j=1}^{N+1} B'_{ij} \vartheta(x_j, t) \quad (11)$$

where N is the number of internal collocation points (i.e. 6) and $A1$ and $B1$ are the matrices of size 7×7 .

Method for finding A , B , A' and B' matrices

The following expressions have been used for calculating A , B , $A1$ and $B1$ matrices.

$$A = CQ^{-1} \quad (12)$$

$$B = DQ^{-1} \quad (13)$$

$$A' = C'Q'^{-1} \quad (14)$$

$$B' = D'Q'^{-1} \quad (15)$$

where Q , C , and D are the matrices of size (14×14) given by

$$Q = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 & \dots & x_1^{N+1} \\ 1 & x_2 & x_2^2 & x_2^3 & \dots & x_2^{N+1} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N+2} & x_{N+2}^2 & x_{N+2}^3 & \dots & x_{N+2}^{N+1} \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 1 & 2x_1 & 3x_1^2 & \dots & (N+1)x_1^N \\ 0 & 1 & 2x_2 & 3x_2^2 & \dots & (N+1)x_2^N \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & 2x_{N+2} & 3x_{N+2}^2 & \dots & (N+1)x_{N+2}^N \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & 0 & 2 & 6x_1 & \dots & N(N+1)x_1^{N-1} \\ 0 & 0 & 2 & 6x_2 & \dots & N(N+1)x_2^{N-1} \\ \cdot & \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & & & & \cdot \\ \cdot & \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & \cdot & & \cdot \\ 0 & 0 & 2 & 6x_{N+2} & \dots & N(N+1)x_{N+2}^{N-1} \end{bmatrix}$$

Q' , C' and D' are the matrices of size (7×7) given by

$$Q' = \begin{bmatrix} 1 & x_1^2 & x_1^4 & \dots & x_1^{2N} \\ 1 & x_2^2 & x_2^4 & \dots & x_2^{2N} \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ 1 & x_{N+1}^2 & x_{N+1}^4 & \dots & x_{N+1}^{2N} \end{bmatrix}$$

$$C' = \begin{bmatrix} 0 & 2x_1 & 4x_1^3 & \dots & 2Nx_1^{2N-1} \\ 0 & 2x_2 & 4x_2^3 & \dots & 2Nx_2^{2N-1} \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ 0 & 2x_{N+1} & 4x_{N+1}^3 & \dots & 2Nx_{N+1}^{2N-1} \end{bmatrix}$$

$$D' = \begin{bmatrix} 0 & 2 & 12x_1^2 & \dots & 2N(2N-1)x_1^{2N-2} \\ 0 & 2 & 12x_2^2 & \dots & 2N(2N-1)x_2^{2N-2} \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdot & & \cdot \\ 0 & 2 & 12x_{N+1}^2 & \dots & 2N(2N-1)x_{N+1}^{2N-2} \end{bmatrix}$$

Discretisation of the governing equation in the bulk phase

The governing Equation (1) has been discretised by using Equations (8) and (9) and is given by

$$\left. \frac{\partial C_b^*}{\partial t} \right|_{x_i} = \frac{-1}{t_{res}} \left[\sum_{j=1}^{N+2} A_{ij} C_b^*(x_j, t) \right] + P \left[\sum_{j=1}^{N+2} B_{ij} C_b^*(x_j, t) \right] - Z(C_b^*(x_i, t) - C_p^*(1, t)) \quad (16)$$

After partial expansion of Equation (16), we obtain

$$\begin{aligned} \left. \frac{\partial C_b^*}{\partial t} \right|_{x_i} = \frac{-1}{t_{\text{res}}} & \left[A_{i,1} C_b^*(0, t) + A_{i,N+2} C_b^*(1, t) + \sum_{j=2}^{N+1} A_{ij} C_b^*(x_j, t) \right] + P[B_{i,1} C_b^*(0, t) \\ & + B_{i,N+2} C_b^*(1, t) + \sum_{j=2}^{N+1} B_{ij} C_b^*(x_j, t) - Z[C_b^*(x_i, t) - C_p^*(1, t)]] \end{aligned} \quad (17)$$

The known conditions at the boundary for Equation (17) are as follows

$$C_b^*(0, t) = 1 \quad (18)$$

$$\text{and } \left. \frac{\partial C_b^*}{\partial z^*} \right|_{z^*=1} = 0 \quad (19)$$

Since $z^* = 1$ represents 14th collocation point, therefore Equation (19) becomes

$$\left. \frac{\partial C_b^*}{\partial z^*} \right|_{x_{14}} = 0 \quad (20)$$

Discretisation of Equation (20) with the help of Equation (8), we get

$$\sum_{j=1}^{N+2} A_{14,j} C_b^*(x_j, t) = 0 \quad (21)$$

After partial expansion of Equation (21), we obtain

$$A_{14,1} C_b^*(0, t) + A_{14,N+2} C_b^*(1, t) + \sum_{j=2}^{N+1} A_{14,j} C_b^*(x_j, t) = 0 \quad (22)$$

$$C_b^*(1, t) = \frac{1}{A_{14,N+2}} \left[-A_{14,1} - \sum_{j=2}^{N+1} A_{14,j} C_b^*(x_j, t) \right] \quad (23)$$

$$\text{Now at } r^* = 1, \quad \frac{\partial C_p^*}{\partial r^*} = R(C_b^* - C_{ps}^*) \text{ where } R = \frac{k_m R_p}{D_e}$$

$$\text{i.e. } \left. \frac{\partial C_p^*}{\partial r^*} \right|_{r^*=1} = R (C_b^* - C_{ps}^*) \quad (24)$$

Since $r^* = 1$ represents 7th collocation point, therefore Equation (24) becomes

$$\left. \frac{\partial C_p^*}{\partial r^*} \right|_{r_7} = R (C_b^* - C_{ps}^*) \quad (25)$$

Discretisation of Equation (25) with the help of Equation (10) gives

$$\sum_{j=1}^{N+1} A'_{7,j} C_p^*(x_j, t) = R (C_b^*(x_i, t) - C_p^*(1, t)) \quad (26)$$

After the partial expansion of Equation (26), we obtain

$$\left[\left(\sum_{j=1}^N A'_{7,j} C_p^*(x_j, t) \right) + A'_{7,N+1} C_p^*(1, t) \right] = R(C_b^*(x_i, t) - C_p^*(1, t)) \quad (27)$$

$$C_p^*(1, t) = \frac{1}{[A'_{7,N+1} + R]} \left[RC_b^*(x_i, t) - \left(\sum_{j=1}^N A'_{7,j} C_p^*(x_j, t) \right) \right] \quad (28)$$

Substituting the values of $C_b^*(0, t)$, $C_b^*(1, t)$ and $C_p^*(1, t)$ from Equations (18), (23) and (28) in Equation (17), we get

$$\begin{aligned} \frac{\partial C_b^*}{\partial t} \Big|_{x_i} &= \frac{-1}{t_{res}} \left[A_{i,1} + \frac{A_{i,N+2}}{A_{14,N+2}} \left\{ -A_{14,1} - \left(\sum_{j=2}^{N+1} A_{14,j} C_b^*(x_j, t) \right) \right\} + \left(\sum_{j=2}^{N+1} A_{i,j} C_b^*(x_j, t) \right) \right] \\ &+ P \left[B_{i,1} + \frac{B_{i,N+2}}{A_{14,N+2}} \left\{ -A_{14,1} - \left(\sum_{j=2}^{N+1} A_{14,j} C_b^*(x_j, t) \right) \right\} + \sum_{j=2}^{N+1} B_{i,j} C_b^*(x_j, t) \right] \\ &- Z \left[C_b^*(x_i, t) - \frac{1}{[A'_{7,N+1} + R]} \left\{ RC_b^*(x_i, t) - \left(\sum_{j=1}^N A'_{7,j} C_p^*(x_j, t) \right) \right\} \right] \end{aligned} \quad (29)$$

$$\begin{aligned} \frac{\partial C_b^*}{\partial t} \Big|_{x_1} &= \frac{-1}{t_{res}} \left[A_{2,1} + \frac{A_{2,14}}{A_{14,14}} \left\{ -A_{14,1} - \left(\sum_{j=1}^{12} A_{14,j+1} C_b^*(x_j, t) \right) \right\} + \left(\sum_{j=1}^{12} A_{i+1,j+1} C_b^*(x_j, t) \right) \right] \\ &+ P \left[B_{2,1} + \frac{B_{2,14}}{A_{14,14}} \left\{ -A_{14,1} - \left(\sum_{j=1}^{12} A_{14,j+1} C_b^*(x_j, t) \right) \right\} + \sum_{j=1}^{12} B_{i+1,j+1} C_b^*(x_j, t) \right] \\ &- Z \left[C_b^*(x_1, t) - \frac{1}{[A'_{7,7} + R]} \left\{ RC_b^*(x_1, t) - \left(\sum_{j=1}^6 A'_{7,j} C_p^*(x_j, t) \right) \right\} \right] \end{aligned} \quad (30)$$

The bulk concentration in the gas phase can be written for 12 internal points. For programming, i and j indices of A and B coefficients of Equation (29) were changed to $i + 1$ and $j + 1$. By substituting $i = 1$ (first collocation point) in Equation (29), (30) is obtained. Hence by substituting different values of i , remaining 11 equations will be obtained at each collocation point along the axial length of the column. These 12 equations obtained will yield solute concentrations at different times in the bulk phase of the bed.

Discretisation of the governing equation inside the pores of the adsorbent

The governing Equation (7) has been discretised by using Equation (10) and (11) and is given by

$$\frac{\partial C_p^*}{\partial t} \Big|_{r_i^*} = \frac{T\alpha (1 + KC_{b0} C_p^*(r_i, t))^2}{\alpha (1 + KC_{b0} C_p^*(r_i, t))^2 + \rho(1 - \alpha)q_m K} \left[\left(\sum_{j=1}^{N+1} B'_{i,j} C_p^*(x_j, t) \right) + \frac{2}{r_i^*} \left(\sum_{j=1}^{N+1} A'_{i,j} C_p^*(x_j, t) \right) \right] \quad (31)$$

After the partial expansion of Equation (31), we obtain

$$\left. \frac{\partial C_p^*}{\partial t} \right|_{r_i^*} = \frac{T\alpha (1 + KC_{b0}C_p^*(r_i, t))^2}{\alpha (1 + KC_{b0}C_p^*(r_i, t))^2 + \rho(1 - \alpha)q_m K} \left[\left\{ B'_{i,N+1} C_p^*(1, t) + \left(\sum_{j=1}^{N+1} B'_{ij} C_p^*(x_j, t) \right) \right\} + \frac{2}{r_i^*} \left\{ A'_{i,N+1} C_p^*(1, t) + \left(\sum_{j=1}^N A'_{ij} C_p^*(x_j, t) \right) \right\} \right] \quad (32)$$

$$\left. \frac{\partial C_p^*}{\partial t} \right|_{r_i^*} = \frac{T\alpha (1 + KC_{b0}C_p^*(r_i, t))^2}{\alpha (1 + KC_{b0}C_p^*(r_i, t))^2 + \rho(1 - \alpha)q_m K} \left[C_p^*(1, t) \left\{ B'_{i,N+1} + \frac{2A'_{i,N+1}}{r_i^*} \right\} + \left(\sum_{j=1}^N B'_{ij} C_p^*(x_j, t) \right) + \frac{2}{r_i^*} \left(\sum_{j=1}^N A'_{ij} C_p^*(x_j, t) \right) \right] \quad (33)$$

Substituting the value of $C_p^*(1, t)$ from Equation (28) in Equation (33) we get

$$\left. \frac{\partial C_p^*}{\partial t} \right|_{r_i^*} = \frac{T\alpha (1 + KC_{b0}C_p^*(r_i, t))^2}{\alpha (1 + KC_{b0}C_p^*(r_i, t))^2 + \rho(1 - \alpha)q_m K} \left[\left\{ \frac{1}{[A'_{7,N+1} + R]} \left[RC_b^*(x_i, t) - \left(\sum_{j=1}^N A'_{7,j} C_p^*(x_j, t) \right) \right] \right\} \left\{ B'_{i,N+1} + \frac{2A'_{i,N+1}}{r_i^*} \right\} + \left(\sum_{j=1}^N B'_{ij} C_p^*(x_j, t) \right) + \frac{2}{r_i^*} \left(\sum_{j=1}^N A'_{ij} C_p^*(x_j, t) \right) \right] \quad (34)$$

6 collocation points have been taken along the radial position of the adsorbent particle. At each collocation point along the radial direction 12 ODE's can be written. Therefore in total 72 ODE's can be obtained from Equation (34). At first collocation point i.e. at r_1^* and at x_1 the following equation will be obtained.

$$\left. \frac{\partial C_p^*}{\partial t} \right|_{r_1^*} = \frac{T\alpha (1 + KC_{b0}C_p^*(r_1, t))^2}{\alpha (1 + KC_{b0}C_p^*(r_1, t))^2 + \rho(1 - \alpha)q_m K} \left[\left\{ \frac{1}{[A'_{7,7} + R]} \left[RC_b^*(x_1, t) - \left(\sum_{j=1}^6 A'_{7,j} C_p^*(x_j, t) \right) \right] \right\} \left\{ B'_{1,7} + \frac{2A'_{1,7}}{r_1^*} \right\} + \left(\sum_{j=1}^6 B'_{1,j} C_p^*(x_j, t) \right) + \frac{2}{r_1^*} \left(\sum_{j=1}^6 A'_{1,j} C_p^*(x_j, t) \right) \right] \quad (35)$$

Hence by substituting different values of i , remaining 71 equations will be obtained at each collocation point along the radial position of the adsorbent particle. These 72 equations obtained will give the variation of solute concentration in the pores with time at various radial positions along the particle and at various axial locations along the bed.

Implementation of a computer program

MATLAB code was written to solve these 84 coupled ODE's, obtained in the previous section by using ODE15s solver. The best fit between experimental and model-predicted values were used to deduce the values of q_m and K . For this purpose, an objective function was used and defined as

the difference between experimental and predicted VOC concentration at the outlet with respect to time. Isqnonlin, a built-in optimisation code of MATLAB, was used to minimise the value of objective function [33]. The computation time was found to be less than a minute on a Pentium IV machine.

Experimental studies

Details regarding experimental set up along with the procedure for carrying out fixed bed adsorption experiments have been discussed here.

Materials and methods

The properties of granular activated carbon are illustrated in Table 1, obtained from Loba Chemie Pvt. Limited, India. Toluene of AR grade was used in the work, procured from Sd fine-Chem Limited, India.

Concentration of toluene in the gas-vapour mixture was measured by employing a gas chromatograph (Netel MICHRO-9100) consisting of SE-30 type capillary column using flame ionisation detector. Following conditions were maintained during analysis: (i) injector temperature: 200°C (ii) oven temperature: 150°C and (iii) detector temperature: 220°C. Gas sampling was done with the help of gas lock syringe of 1 mL capacity (Hamilton-Bonaduz).

Experimental set-up

Figure 1 demonstrates the schematic representation of experimental set-up. Firstly, to prepare dilute gas-vapour mixture, nitrogen gas was made to bubble through liquid toluene filled up to a certain level in a 0.7 m long water-jacketed vertical glass column of 0.05 m diameter. The exiting gas vapour mixture from the glass column was then directed to the mixing chamber where it was mixed with pure nitrogen to get desired toluene concentration following which it was fed to the adsorption column, filled with granular activated carbon [34]. The effluent gas from the adsorption column was then sent to GC for analysis. Breakthrough times were found out in all the experiments, defined as the time in which toluene concentration at the exit of the adsorption column reaches 5% of the inlet toluene concentration.

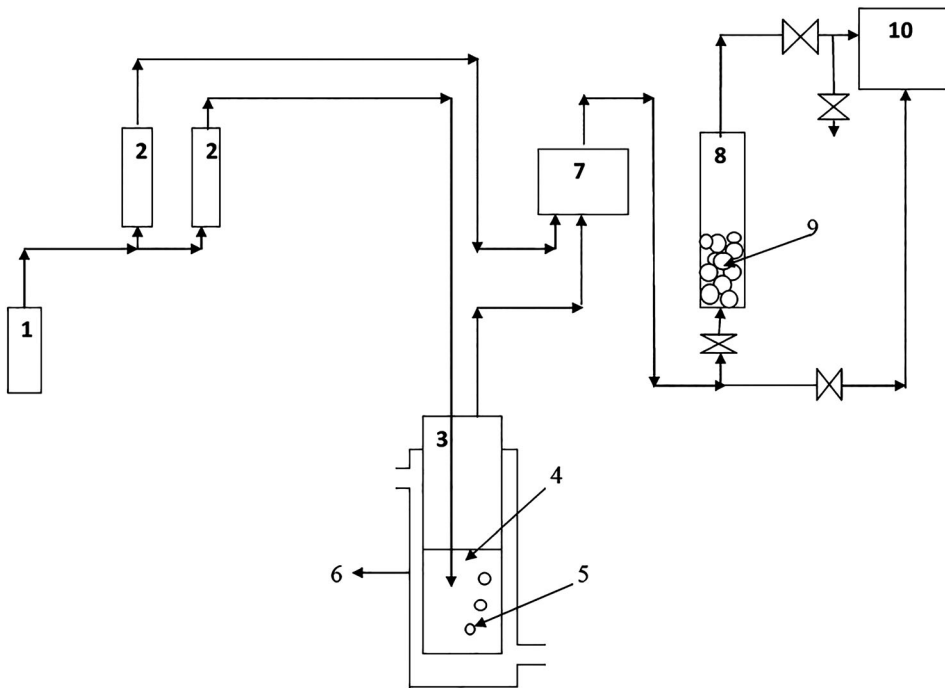
Results and discussion

Effect of bed height

Effect of bed height on breakthrough time was found out for both toluene by conducting the experiments at varying bed heights: 0.015 and 0.025 m, shown in Figure 2. Breakthrough time increased from 5748 to 12175 s (Figure 2) with an increase in bed height from 0.015 to 0.025 m at 11500 ppm inlet gas concentration and 70 mL min⁻¹ gas flow rate. Langmuir isotherm parameters, q_m and K , reported in Table 2, for toluene were found out for the best fit between experimental and model results.

Table 1. Characteristics of activated carbon.

Parameters	Value
Particle size [mm]	1.5
BET surface area [m ² g ⁻¹]	564
Pore volume [cm ³ g ⁻¹]	0.32
Average pore diameter [Å]	22.68
Particle density [kgm ⁻³]	400



1. Nitrogen gas cylinder
2. Rotameters
3. VOC glass column with bubbler
4. Liquid VOC level
5. Gas bubbles rising through liquid VOC
6. Jacket for maintaining isothermal conditions
7. Mixing chamber
8. Adsorption column
9. Adsorbent
10. Gas chromatography with FID

Figure 1. Schematic diagram of the experimental set-up for VOC adsorption.

Effect of inlet concentration

Effect of inlet VOC concentration on breakthrough time was found out by conducting the experiments at varying inlet toluene concentrations: 7000 and 11,500 ppm, shown in [Figure 3](#). Breakthrough time decreased from 6158 to 5748 s ([Figure 3](#)) with the increase in inlet concentration from 7000 to 11,500 ppm at 70 mL min^{-1} gas flow rate and 4 g adsorbent. Langmuir isotherm parameters, q_m and K , reported in [Table 3](#), for toluene were found out for the best fit between experimental and model results.

Slight variations were observed in the values of q_m and K for toluene listed in [Tables 2 and 3](#) owing to which their average values were used in the simulation studies and are listed in [Table 4](#).

Simulation studies

Simulations were further carried out to obtain breakthrough curves under varying operating conditions where experimentation was difficult. Also, the most sensitive parameter was ascertained with respect to breakthrough time by carrying out sensitivity analysis. Sensitivity is defined as the

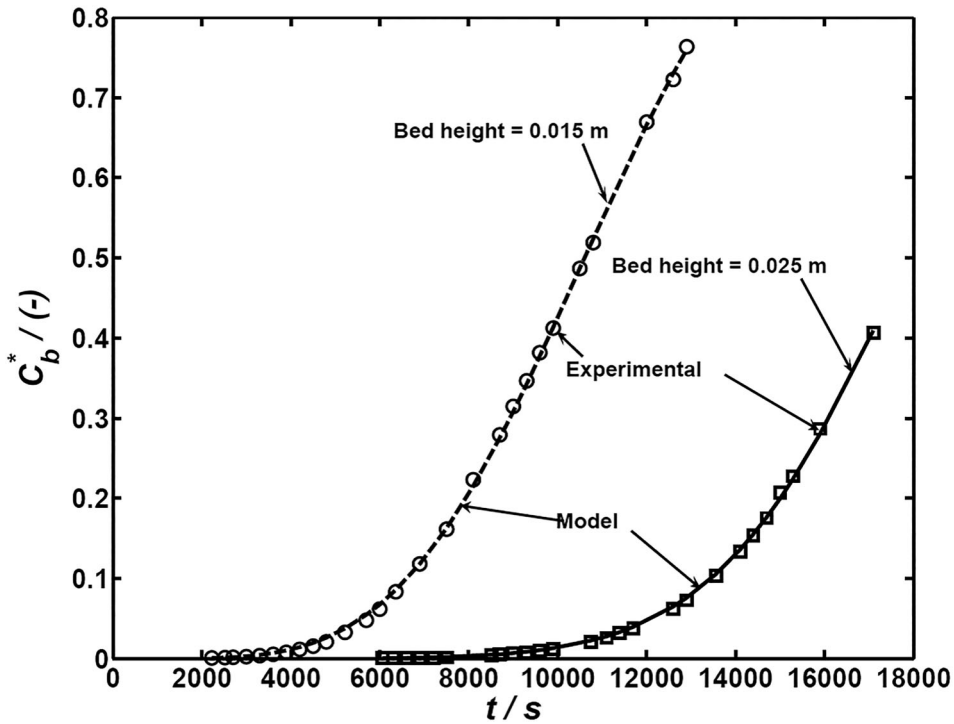


Figure 2. Effect of bed height on breakthrough curve for toluene (Gas flow rate = 70 mL min⁻¹, Inlet concentration = 11,500 ppm, Particle diameter = 0.0015 m).

Table 2. Values of adjustable parameters for various bed heights for toluene.

Parameter	(Gas flow rate = 70 mL min ⁻¹ , Inlet concentration = 11,500 ppm, Particle diameter = 0.0015 m)	
	Bed height = 0.015 m	Bed height = 0.025 m
q_m [kg kg ⁻¹]	0.118	0.111
K [m ³ kg ⁻¹]	34.263	35.696

ratio of change in percentage of the output value (breakthrough time) to the change in percentage of the input value (flow rate, inlet concentration, bed height, particle diameter and effective diffusivity) with respect to their reference values [35]. The reference parameters, listed in Table 5, were selected on the basis of conditions on which experiments were performed and 50% deviation (both positive and negative) with respect to reference value was considered for each parameter. Table 6 provides the values of various parameters for toluene under varying operating conditions essential for carrying out simulation studies. Following expressions were utilised to estimate the values of axial dispersion coefficient, external mass transfer coefficient, and effective diffusivity.

Axial dispersion coefficient, D_L :

$$D_L = \frac{D_p \cdot \mu}{\rho_g \cdot Pe} \quad (36)$$

$$\text{and } Pe = \frac{0.3}{(Re)(Sc)} + \frac{0.5}{1 + (3.8/Re \cdot Sc)}, \quad 0.008 < Re < 400 \text{ and } 0.28 < Sc < 2. \quad (37)$$

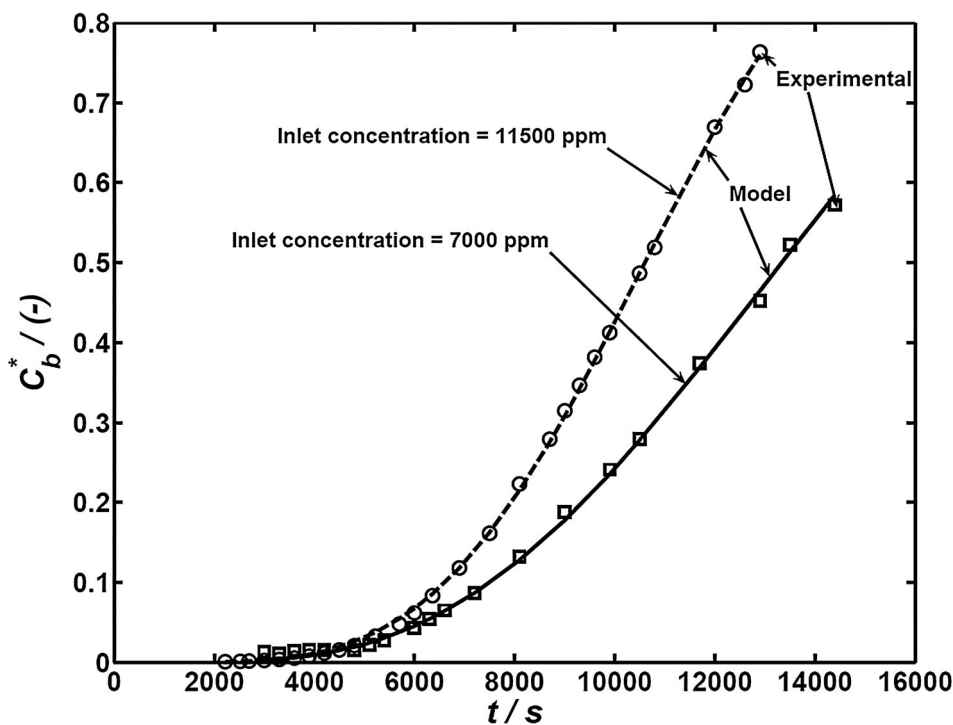


Figure 3. Effect of inlet concentration on breakthrough curve for toluene (Gas flow rate = 70 mL min⁻¹, Bed height = 0.015 m, Particle diameter = 0.0015 m).

Table 3. Values of adjustable parameters for various inlet toluene concentrations.

Parameter	(Gas flow rate = 70 mL min ⁻¹ , Bed height = 0.015 m, Particle diameter = 0.0015 m)	
	Inlet gas concentration = 7000 ppm	Inlet gas concentration = 11500 ppm
q_m [kg kg ⁻¹]	0.118	0.118
K [m ³ kg ⁻¹]	32.951	34.263

Table 4. Langmuir isotherm parameters for toluene.

Langmuir isotherm parameters	Toluene
q_m [kg kg ⁻¹]	0.116
K [m ³ kg ⁻¹]	34.654

Table 5. Reference parameters for toluene.

Parameter	Toluene
Inlet concentration [ppm]	7000
Gas flow rate [mL min ⁻¹]	70
Bed height [m]	0.015
Particle diameter [m]	0.0015
Effective diffusivity [m ² s ⁻¹]	2.2 × 10 ⁻⁸

Table 6. Calculated values of Re, Sc, Pe, D_L , k_m and D_e under various operating conditions for toluene.

S.No.	Q [mL min ⁻¹]	d_p [m]	$U \times 10^4$ [ms ⁻¹]	Re	Sc	Pe	$D_L \times 10^8$ [m ² s ⁻¹]	k_m [ms ⁻¹]	$De \times 10^8$ [m ² s ⁻¹]
1	35	0.0015	2.972	0.0335	1.4788	0.1651	12.1	0.0011	2.2
2	70	0.0015	5.944	0.067	1.4788	0.3291	6.062	0.00149	2.2
3	105	0.0015	8.917	0.1006	1.4788	0.4912	4.06	0.0018	2.2
4	70	0.00075	5.944	0.033	1.4788	0.16507	6.042	0.00214	2.2
5	70	0.00225	5.944	0.1006	1.4788	0.4911	6.092	0.00122	2.2
6	700	0.0015	0.0059	0.6705	1.4788	2.4628	0.81	0.00464	2.2

External mass transfer coefficient, k_m :

$$Sh = 0.91\varphi(Re)^{0.49}(Sc)^{0.33}, \quad 0.01 < Re < 50 \quad (38)$$

$$\text{and } Sh = \frac{k_m D_p}{D_{AB}} \quad (39)$$

Effective diffusivity, D_e :

$$D_e = \frac{\alpha D}{\tau} \quad (40)$$

$$\frac{1}{D} = \frac{1}{D_k} + \frac{1}{D_{AB}} \quad (41)$$

where D_{AB} and D_k can be found out by the following expressions:

$$D_{AB} = \frac{10^{-4} \left(1.084 - 0.249 \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{0.5} \right) T^{1.5} \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{0.5}}{P(r_{AB})^2 f \left(\frac{kT}{\epsilon_{AB}} \right)} \quad (42)$$

$$D_k = 97r_p \left(\frac{T}{M_A} \right)^{0.5} \quad (43)$$

Effect of flow rate

Figure 4 describes the behaviour of breakthrough curve at varying gas flow rates for toluene. Adsorption time was found to be decreasing with rise in flow rate. Since higher flow rates corresponds to enhanced external mass transfer coefficients which in turn leads to greater adsorption rates, eventually resulted in lesser breakthrough times.

Effect of particle size

Effect of particle diameter for toluene on breakthrough curve is presented in Figure 5. Notable increase in breakthrough time was observed with a decrease in particle size. Particularly for small size particles, this perceptible attribute of breakthrough curve results into lesser time for the bed to get saturated owing to sudden rise in VOC concentration following the break-through of the bed. On the contrary, for large size particles, longer diffusion path inside the pores results in to lesser breakthrough time and hence greater time for the saturation of the bed.

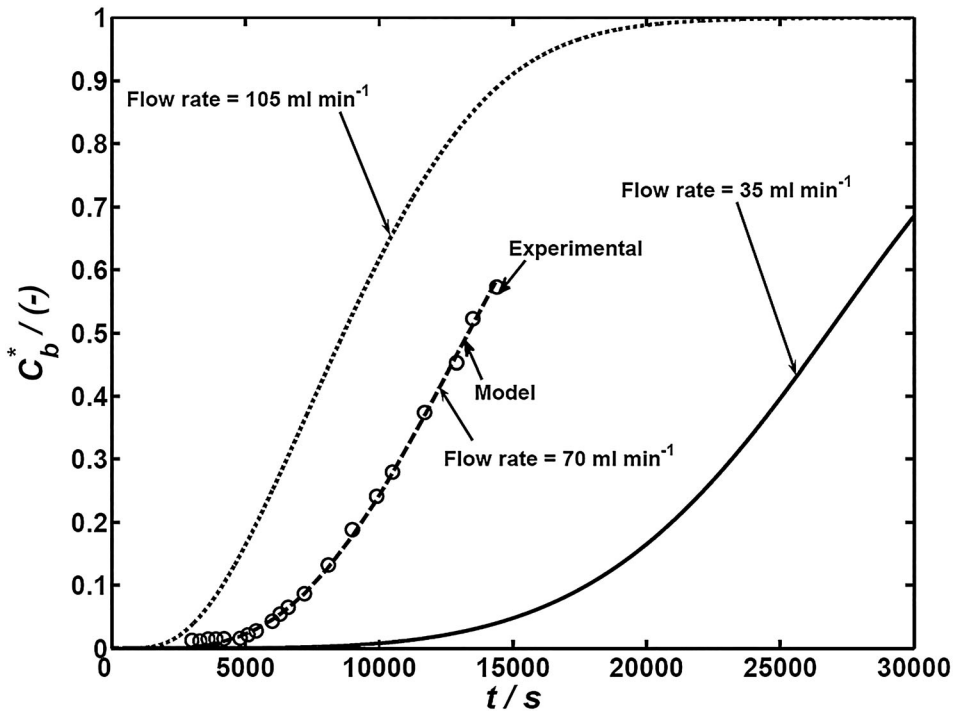


Figure 4. Effect of gas flow rate on breakthrough curve for toluene (Inlet concentration = 7000 ppm, Bed height = 0.015 m, Particle Diameter = 0.0015 m, Effective diffusivity = $2.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$).

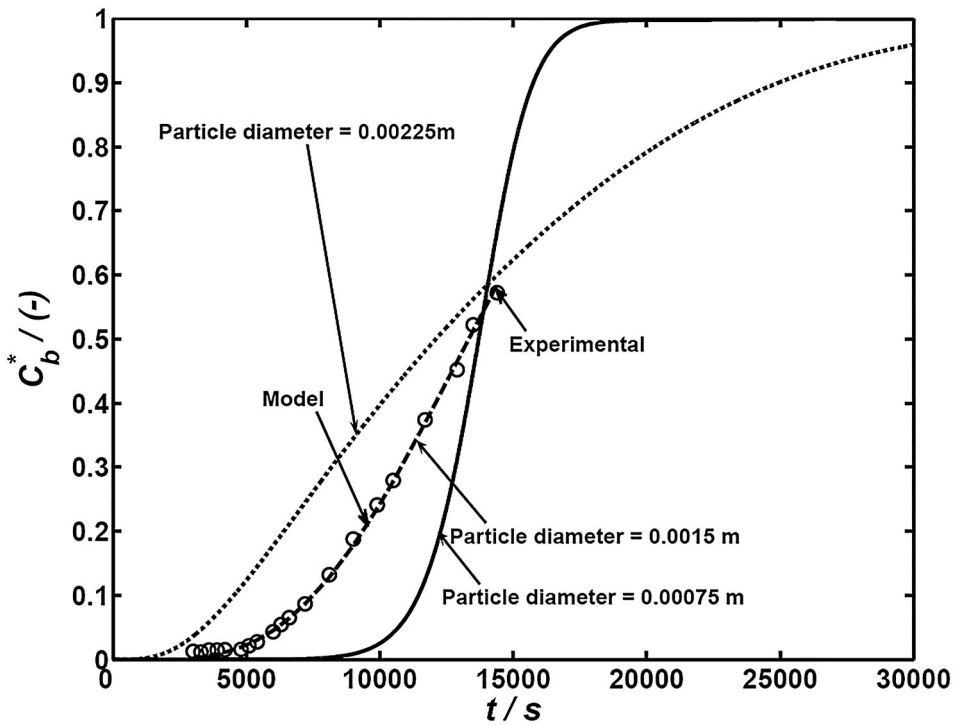


Figure 5. Effect of particle diameter on breakthrough curve for toluene (Inlet concentration = 7000 ppm, Gas flow rate = 70 mL min⁻¹, Bed height = 0.015 m, Effective diffusivity = $2.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$).

Effect of pore diffusivity

The response of breakthrough curves for toluene at different pore diffusivities is illustrated in Figures 6 and 7 at varying flow rates (70 and 700 mL min⁻¹). At a given flow rate, the breakthrough time was found to be increasing with the increase in pore diffusivity but the saturation time decreased. It was also observed that rear end has a longer tail when simulations were carried out for low values of pore diffusivity implying higher internal resistance owing to which time taken by the bed to get saturated was found to be more. But on the other hand, breakthrough comes earlier because the total amount to be adsorbed per gram of adsorbent at saturation will remain same at identical flow rates and inlet concentrations for different pore diffusivities.

On comparing Figures 6 and 7, at higher flow rates (700 mL min⁻¹) adsorption process is found to be pore diffusion limited because of almost 600% change in breakthrough time in comparison to only 85% change in breakthrough time at lower flow rates (70 mL min⁻¹) was observed when pore diffusivity was varied from 1.1×10^{-8} – 3.3×10^{-8} m² s⁻¹. At low flow rates, the adsorption process is external mass transfer controlled.

Effect of bed height

Figure 8 shows the pattern of breakthrough curve when simulations were conducted at varying bed heights for toluene by keeping inlet concentration and flow rates constant. Here breakthrough time was found to be increasing with the increase in bed height. This increase in breakthrough time is due to the fact that more amount of adsorbent material will be packed in a column at greater bed heights resulting in delayed breakthrough.

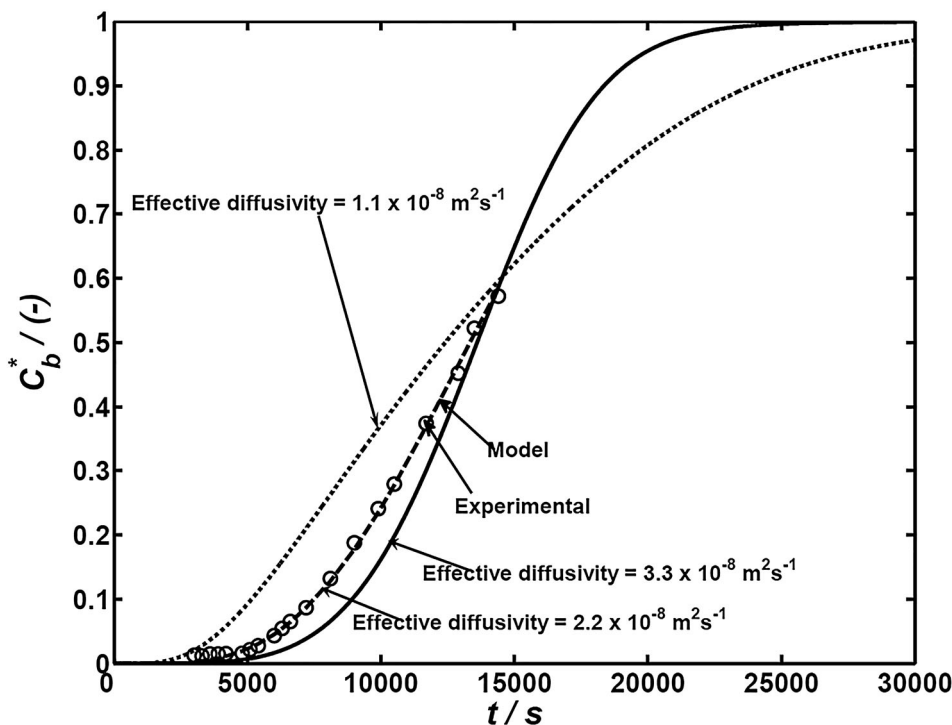


Figure 6. Effect of pore diffusivity on breakthrough curve for toluene (Inlet concentration = 7000 ppm, Gas flow rate = 70 mL min⁻¹, Bed height = 0.015 m, Particle Diameter = 0.0015 m).

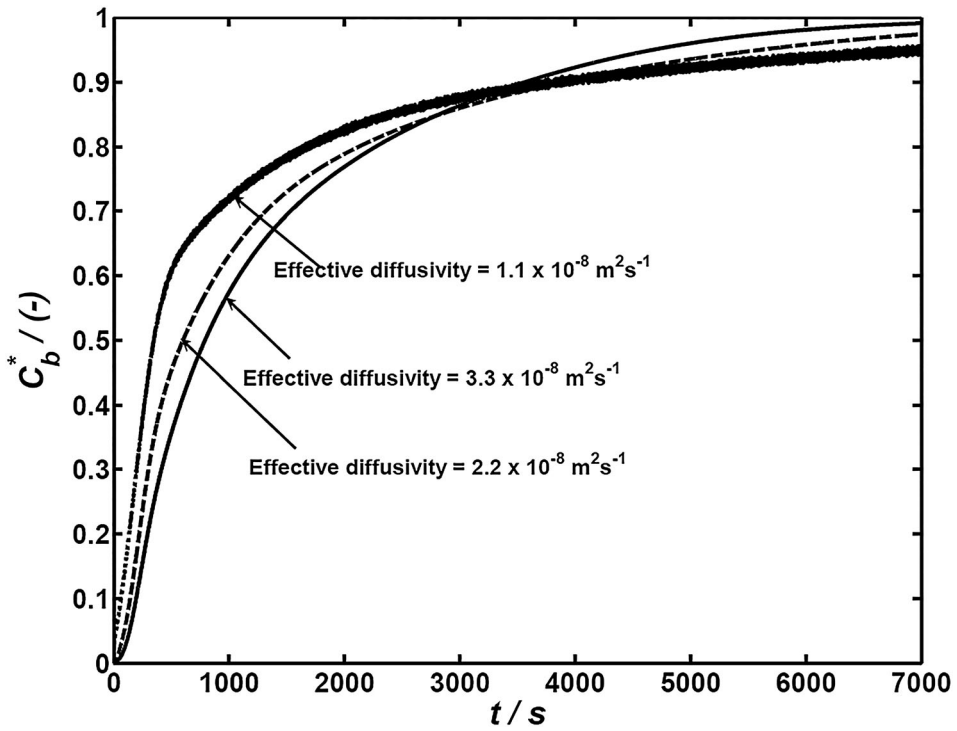


Figure 7. Effect of pore diffusivity on breakthrough curve for toluene (Inlet concentration = 7000 ppm, Gas flow rate = 700 mL min⁻¹, Bed height = 0.015 m, Particle Diameter = 0.0015 m).

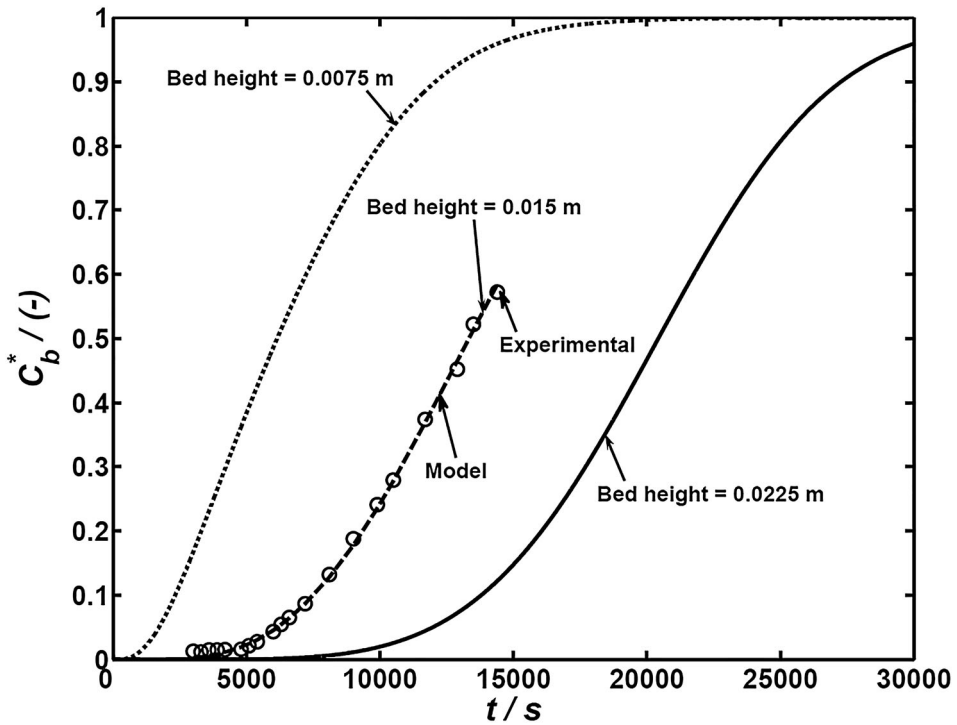


Figure 8. Effect of bed height on breakthrough curve for toluene (Inlet concentration = 7000 ppm, Gas flow rate = 70 mL min⁻¹, Particle Diameter = 0.0015 m, Effective diffusivity = $2.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$).

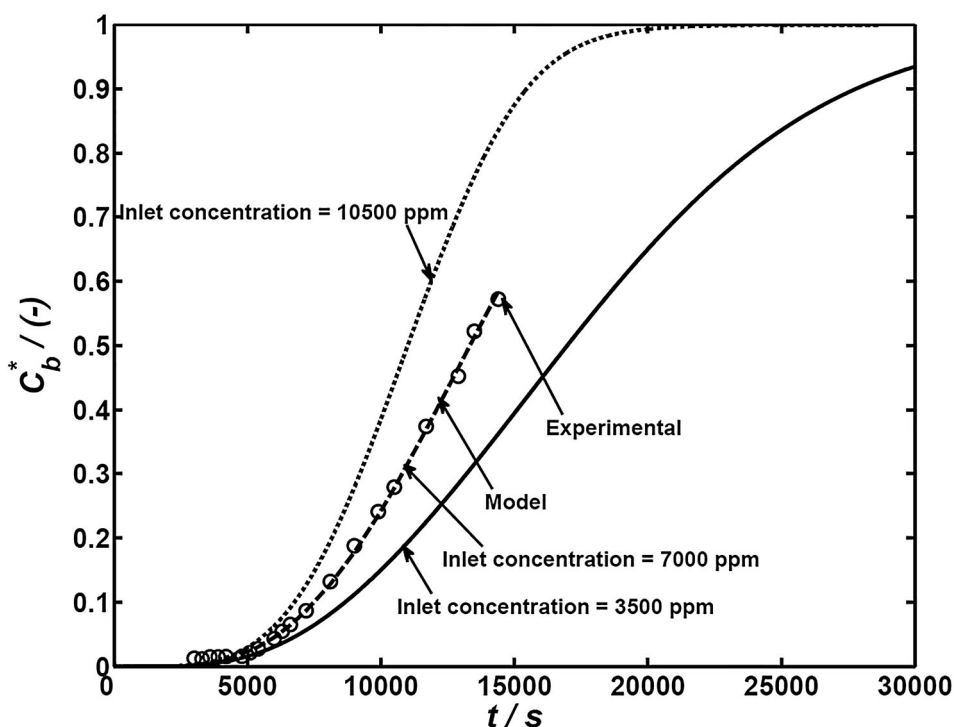


Figure 9. Effect of inlet concentration on breakthrough curve for toluene (Gas flow rate = 70 mL min^{-1} , Bed height = 0.015 m , Particle Diameter = 0.0015 m , Effective diffusivity = $2.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$).

Table 7. Sensitivity analysis.

Parameters	Change in breakthrough time
$Q(+)$	-0.93
$Q(-)$	-2.92
$D_e(+)$	0.479
$D_e(-)$	0.7
$d_p(+)$	-0.896
$d_p(-)$	-1.475
$L(+)$	1.891
$L(-)$	1.439
$C_{b0}(+)$	-0.154
$C_{b0}(-)$	-0.253

Effect of inlet concentration

Figure 9 elucidates the effect of toluene inlet concentration on the breakthrough curve. Breakthrough time was found to decrease with the increase in inlet toluene concentration and consequently total saturation time was also found to be lesser at higher inlet concentrations. It may be explained by taking into account the shape of a favourable adsorption isotherm which indicates higher velocity of mass transfer zone at higher concentrations owing to lesser slope of the isotherm in this region (high concentration).

Sensitivity analysis

Here sensitivity analysis was performed by calculating the change in breakthrough time for any parameter from the following expression:

$$\text{Change in breakthrough time} = \frac{\% \text{ change in breakthrough time w.r.t reference value}}{\% \text{ change in respective parameter w.r.t reference parameter}} \quad (44)$$

The results obtained are summarised in Table 7. The negative value implies a decrease in breakthrough time when the value of that parameter is increased (or vice versa). Gas flow rate was found to be most sensitive parameter and bed height was the second most sensitive parameter regarding breakthrough time for both toluene. The effect of other parameters on breakthrough time was found to be insignificant in comparison to flow rate and bed height.

Conclusions

The removal of toluene by adsorption on granular activated carbon was studied both experimentally as well as theoretically and following conclusions have been drawn:

- (1) Adsorption technique is found to be effective in removing VOCs from the dilute streams as breakthrough time considerably decreased with increase in VOC concentration.
- (2) Greater bed heights are desirable to achieve breakthrough in a longer time.
- (3) Simulation studies and hence sensitivity analysis divulge flow rate to be the most sensitive parameter for the range studied, implying adsorption to be externally mass transfer controlled.

Disclosure statement

No potential conflict of interest was reported by the author.

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